

BSDS (2025 Semester II): Statistics II

Midsem: Part B

NAME :	ROLL :
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Time: 60 minutes

Total marks: 26

Answer all questions.

1. Let $X_1, \dots, X_n$ be n i.i.d. samples from the population $F_{\boldsymbol{\theta}}$ with mean $\mu_{\boldsymbol{\theta}}$ and finite variance $\sigma_{\boldsymbol{\theta}}^2$, and let $\bar{X}_n = n^{-1} \sum_{i=1}^n X_i$. Show that $E(\bar{X}_n) = \mu_{\boldsymbol{\theta}}$ and $\text{var}(\bar{X}_n) = n^{-1} \sigma_{\boldsymbol{\theta}}^2$. [2 + 3]

2. let $X_1, \dots, X_n$ be a random sample from the log-normal distribution with parameters $\boldsymbol{\theta} = (\mu, \sigma^2)$ and p.d.f.

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma x} \exp \left\{ -\frac{1}{2\sigma^2} (\log x - \mu)^2 \right\}, \quad \text{for } x > 0, \mu \in \mathbb{R}, \sigma^2 > 0.$$

Find the MLE of $\boldsymbol{\theta}$.

[8]

3. Find the UMVUE of the parameters μ and σ^2 for log-normal distribution (described in Q.2). [8]

4. Compare the MLE and UMVUE estimators of σ^2 for the log-normal distribution.

[5]

Rough work

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